



Explainable deep learning approach to predict residential electricity demand

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Abstract The escalating demand for electricity in residential buildings is a significant concern that necessitates a comprehensive understanding of the underlying reasons and responsible parameters. The electricity forecasting problem involves integrating the time series of weather conditions and electricity consumption into an intelligent model that can explain the influence of weather parameters on electricity consumption. Artificial intelligence techniques have shown excellent electricity prediction performance, but the crucial challenge is reliable and interpretable predictions. This paper proposes an intelligent electricity prediction approach by exploiting the potential of eXplainable Artificial Intelligence (XAI). The proposed electricity demand prediction model integrates a transformer-based long short-term memory model with genetic algorithms. Further, an XAI tool has been applied to interpret the prediction results and provide a deeper understanding of the factors influencing electricity consumption. Two experiments have been

conducted to evaluate predictive performance, the first on a dataset of real-time electricity consumption and the second on a benchmark dataset of residential buildings. The proposed approach outperformed the other state-of-the-art models and achieved the lowest mean absolute error.

Keywords Electricity prediction · Optimization · Deep learning · EXplainable AI · Residential buildings

1 Introduction

Household electricity consumption is increasing swiftly; that accounts for approximately a quarter of total energy utilized. This alarming rise in energy usage is a top priority for energy suppliers and utility companies. With the advent of artificial Intelligence (AI) and machine (ML) techniques, several researchers considered energy prediction a sustainable solution (Chabane et al. 2024; Kaur et al. 2022). ML models are opaque and can be considered black box models because their inner workings and decision-making are not interpretable and understandable by the users (Zhang et al. 2020). In critical applications such as electricity forecasting, the stakeholders have to be confident about their predictions and the factors responsible for such predictions. Explainable AI (XAI) is an emerging research area that provides meaningful insights and more informed decisions to improve predictive performance.

Most of the research papers have explored the application of XAI in image and text processing, and very little work has been done using time-series data. Explainable AI has been utilized in areas such as photovoltaic power generation (Kuzlu et al. 2020), ride-hailing demand prediction (Li 2022), and power system emergency control (Zhang et al. 2020). The researchers have deployed XAI techniques in electricity

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forecasting to enhance the interpretability, transparency, and explainability of ML models (Henriksen et al. 2022; Mohanty et al. 2022). The Local Interpretable Model Agnostic Explanations (LIME) XAI tool can be applied to explain the individual predictions of regression and classification models (Ribeiro et al. 2016). Shapley Additive exPlanations (SHAP) are helpful in explaining all the model predictions. EL15 is another Python-based XAI tool that helps to interpret various regression and classification models (Kuzlu et al. 2020).

The present work leverages XAI method to identify the influential features that impact the accuracy of electricity predictions. To forecast and optimize daily energy demand a genetic algorithm (GA) has been integrated with long short-term memory (LSTM) model. The efficiency of the proposed explainable electricity prediction and optimization method has been validated using real-time and state-of-the-art datasets of electricity consumption. The unique contributions of this research work are mentioned below.

1.1 Our contributions

- A hybrid GA-LSTM-based prediction and optimization approach has been developed for residential buildings by considering the climatic conditions and time-based features. LSTM model employs a transformer layer during model implementation to handle long-term time dependencies. GA optimization approach has been adopted to compute the optimal hyperparameter of the LSTM model, namely input window size, hidden layers, learning rate, and number of neuron.
- The XAI tool based on SHapely Additive exPlanation (SHAP) has been applied for interpreting and identifying the responsible factors for electricity consumption. The impact of climatic conditions and temporal features on energy consumption patterns is investigated.
- The real-time electricity consumption dataset and Indian BuILdings Energy coNsumption Dataset (I-BLEND) of residential buildings have been exploited to validate the proposed XAI-based GA-LSTM electricity prediction and optimization approach.

The remaining paper is structured in the following way: Sect. 2 discussed the literature work in the field of prediction and XAI approaches. Further, Sect. 3 presents the proposed methodology, and Sect. 4 elaborates the prediction and explainability results. Lastly, Sect. 5 concludes the proposed approach.

2 Literature review

The relevant literature on energy prediction and optimization approaches has been studied and reviewed in the context of

residential and other types of buildings. The explainable AI techniques deployed by researchers for intelligent energy prediction have also been discussed. An improved energy optimization approach based on genetic algorithm (GA) and PSO has been proposed by Ullah and Kim (2017) to minimize energy consumption with the maximum user comfort level. While experimentation, visual/thermal comfort and air quality were taken as comfort parameters for optimizing energy consumption. Hafeez et al. (2020) have proposed a Genetic Wind-Driven (GWDO) algorithm for a day and week-ahead load forecasting model. The effectiveness of the proposed model has been demonstrated using historical load data of USA power grids. Arora et al. (2024) proposed structural modeling for the use of energy-efficient appliances and investigated the purchasing behavior of consumers for these appliances. This study revealed the factors influencing the consumer's intention of using energy-efficient appliances.

A neural network model; Multi-Layer Perceptron (MLP) has been integrated with Artificial Bee Colony (ABC) and Particle Swarm Optimization (PSO) to improve the prediction of the heating and cooling loads by Zhou et al. (2020). The experiment was performed using a residential building dataset by Tsanas and Xifara (2012). An improved sparrow search algorithm (ISSA) for energy consumption optimization has been proposed by Xue et al. (2023) to solve the optimal chiller loading problem. The chaotic circle mapping has been used to improve the initial solution quality of the optimization algorithm. The simulation results showed more energy savings, low running time, and a fast convergence rate using ISSA. Grey wolf optimization (GWO) algorithm (Li et al. 2021) has been integrated with a back-propagation neural network model for predicting and optimizing the indoor thermal temperature level and energy consumption.

An optimization approach using GA has been proposed utility companies to reduce the electricity load for heating and air conditioning (Arabali et al. 2012). Tran and Nguyen (2022) investigated an energy cost minimization problem under uncertain conditions. Mixed-integer linear programming model has been deployed to find the optimal energy plans for customers. Ngo et al. (2022) proposed SVR and gray wolf optimization-based 48-step ahead energy prediction approach for non-residential buildings. A neural network model has been integrated by Sebi (2022) with a meta-heuristic weight optimization algorithm for solar irradiance prediction. Kaur et al. (2023) integrated LSTM model with GA to forecast energy usage of individual home appliances. GA is deployed for hyperparameter selection of the LSTM model. Cai et al. (2023) has implemented a wind power forecasting approach using GWO and the LSTM model. They deployed climate data and extracted different time scale features like day, month, season, and annual to analyze wind power characteristics. Barua et al. (2024) proposed an efficient and secure energy consumption system for IoT based

smart homes. Kuzlu et al. (2020) deployed RF model to estimate solar photovoltaic power generation. The XAI methods SHAP, LIME and EL15, have been utilized to determine the feature contribution in the prediction results.

Zhang et al. (2020) deployed the XAI-based SHAP method to interpret the deep reinforcement learning model for improving the human-computer interaction and machine learning models. Henriksen et al. (2022) proposed an XAI-based deep neural networks model for electricity consumption prediction of Norwegian residential buildings. The authors have applied and compared SHAP and LIME tools for model explainability and analyzed that SHAP provides both local and global explanations. Mohanty et al. (2022) deployed XAI tools such as LIME and SHAP to improve the interpretation of the prediction model and determine the factors influencing energy demand in households. Kruse et al. (2022) proposed an explainable machine learning model using gradient-boosted trees and SHAP method to predict renewable power consumption. XAI method has extracted the features influencing the performance of the forecasting model. Bhandary et al. (2024) implemented LIME and SHAP tools to identify the important features contributing to electricity prediction and also analyzed the external factors affecting electricity consumption patterns in households.

The comprehensive literature review emphasized the latest energy prediction and optimization techniques, a comparative analysis of various prediction, optimization, and XAI approaches is presented in Table 1. Subsequently, the critical analysis presents the significant contribution and limitations of the existing works. Several authors have adopted machine learning and deep learning models for predicting electricity consumption. However, in deep learning models, choosing the optimal hyperparameters is crucial for getting accurate energy predictions. Therefore, some authors have deployed optimization techniques for selecting hyperparameters to improve the predictive performance of deep learning algorithms. Moreover, a few authors (Mohanty et al. 2022; Elayan et al. 2022; Chakraborty and Datta 2017) have developed explainable machine learning models to enhance the prediction's understandability. Kuzlu et al. (2020) highlighted that LIME may not fairly produce global model explanations. There are various limitations and challenges in the current explainability-based prediction approaches that need to be addressed. So, the motivation behind this research work is to estimate electricity demand accurately and offer comprehensive explanations of the prediction outcomes. The deep insights into the influential parameters will improve prediction performance while allowing a profound understanding of consumption patterns.

2.1 Research gaps

The extensive literature review has found various weaknesses and gaps in the existing works, which are mentioned below:

- A limited amount of work has been done using real-time data of energy usage behavior that should be collected and integrated with energy data to understand the effect of behavior on energy consumption.
- An optimal and effective hyperparameter selection method is required to balance the performance and computational complexity.
- Few authors have integrated prediction models with XAI techniques to provide actionable insights or recommendations.
- Some authors have focused on climatic conditions and occupancy behavior of users for implementing accurate energy prediction models.

3 Proposed methodology

This paper proposed an explainable prediction and optimization approach for estimating electricity consumption in residential buildings. The architecture of the proposed work is shown in Fig. 1, depicting the use of explainable AI in electricity prediction and optimization. This framework consists of data collection, an AI-based prediction model, an optimization approach, and an explainable AI model. The following subsections elaborate on these modules in detail.

3.1 Dataset description and preprocessing

The first stage of the proposed methodology is to collect real-time electricity consumption of residential buildings from Punjab State Power Corporation Limited (PSPCL), Punjab, India. This dataset consists of smart meters' power consumption (kWh) readings. The weather dataset (RP5 Weather Archive 2023) of Chandigarh International Airport station is used to characterize the variable climatic conditions, including features temperature, humidity, and wind speed. A benchmark dataset of buildings has been exploited to validate the predictive performance of the proposed work. It is a 52-month public dataset; Indian Buildings Energy consumption Dataset (I-BLEND) (Rashid et al. 2019). It is collected from commercial and residential buildings of an educational institute at a one-minute frequency. The parameters of the energy dataset are voltage, current, power, frequency, and power factor. The parameters of the weather dataset include temperature, humidity, wind speed, and wind direction, recorded at 30-minute intervals.

Table 1 Comparison of state-of-the-art prediction, optimization and XAI approaches

References	Year	Techniques/algorithms used	Optimization parameters	Focus area/strengths
Ullah and Kim (2017)	2017	Kalman filter, PSO, GA	Temperature, illumination, indoor air quality	To reduce energy consumption, improve comfort index
Muralitharan et al. (2018)	2018	CNN, GA, PSO	CNN hyperparameter	Prediction and optimization of energy demand and supply
Goudarzi et al. (2019)	2019	SVR, FNN, PSO	SVR hyperparameter C, ϵ , kernel constant ρ	Accurate power consumption prediction in intelligent buildings
Kaur and Bala (2019)	2019	ANN, PSO, ABC	ANN hyperparameters	Energy prediction and optimization
Zhou et al. (2020)	2020	MLP, PSO, ABC	Relative compactness, surface area, wall area, roof area, height, orientation, glazing area	Optimization of heating and cooling load
Luo et al. (2020)	2020	LSTM, GA	Number of neurons, dropping rate, learning rate	Building energy management and estimated hourly energy consumption
Hafeez et al. (2020)	2020	Factored conditional restricted Boltzmann machine (FCRBM), Genetic Wind-Driven Optimization Algorithm	Feature selection, hyperparameters	Short-term electricity demand prediction
Chen et al. (2021)	2021	Least Square Support Vector Machine (LSSVM), NSGA-II	Window, roof and wall U-value, window-to-wall ratio, wall and roof solar absorptance	Optimize energy consumption, improve thermal comfort
Chakraborty et al. (2021)	2021	XGBoost, SHAP	Cooling energy consumption, temperature	Explainable energy predictions
Etemad et al. (2022)	2022	ANN, MOGA	Cooling setpoint, outdoor airflow, insulation, window solar transmittance, discomfort glaring index	Optimization of energy consumption, thermal comfort, carbon emission
Hosamo et al. (2022)	2022	GLSVM, NSGA-II	HVAC setpoint, shedding parameters, lighting, air infiltration	Energy consumption optimization and thermal comfort
Mohanty et al. (2022)	2022	RF, SHAP	Energy use, temperature, humidity, dew point, pressure, wind speed	XAI application to predict energy consumption
Huang et al. (2022)	2022	SVM, GWO, point estimation method	Renewable energy system in smart buildings	Predicted output power of wind turbines and photovoltaics
Zhao et al. (2022)	2022	Autoregressive Moving Average (ARMA), Generalized Autoregressive Conditional Heteroscedasticity (GARCH)	Australian electricity data	Electricity price prediction
Elayan et al. (2022)	2023	LSTM, IBM AI Explainability 360	Power consumption, day, month	Internet of Behavior (IOB) concept and XAI model to reduce energy consumption
Sakkas et al. (2023)	2023	Genetic programming, SHAP	Real-time electricity data, Greece	Electricity prediction
Bhandary et al. (2024)	2024	XGBoost, LSTM, RF, LIME, SHAP	Real-time electricity data of apartment, Bangalore, India	Electricity demand prediction
Janjua et al. (2024)	2024	XGB, LSTM, SHAP	Smart Grid, Paris, France	Electricity forecasting

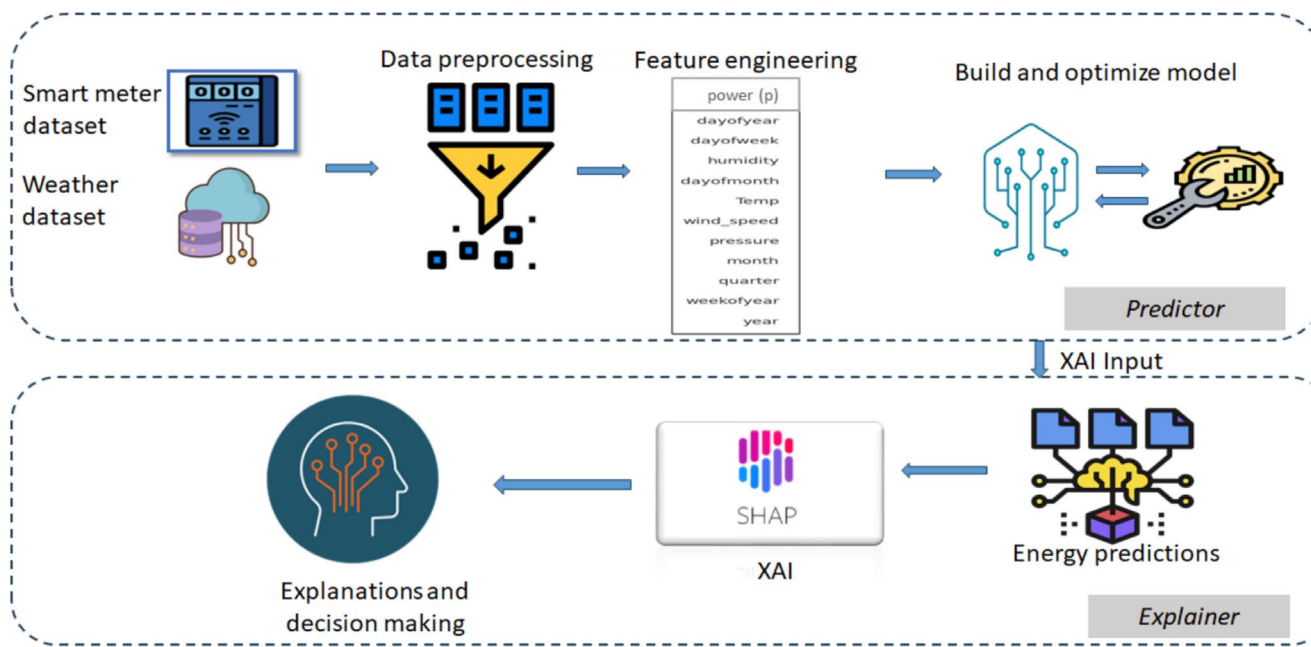


Fig. 1 Architecture of explainable AI-based electricity prediction and optimization approach

Data preprocessing is necessary prior to training a deep learning model due to its significant impact on prediction accuracy. The electricity consumption dataset may have some missing values, which are handled by data preprocessing procedures for the electricity dataset.

- The input dataset of electricity consumption contains missing and NULL values identified during data exploration phase and these are treated using linear interpolation. This technique best suits time series problems due to its ability to handle unknown values by preserving the continuity of the data. The missing values are determined using linear interpolation that computes the average of past and present measurements to fill the missing values.
- Feature normalization has been applied to electricity consumption measurements to ensure the same feature scale throughout the dataset. In this paper, the Min–Max scalar is adapted to convert the electricity consumptions into the range of 0 and 1 (Pedregosa et al. 2011). The input feature, power, of dataset P_t , after normalization, new dataset P_n is given by Eq. 1:

$$P_n = \frac{P_t - \min(P_t)}{\max(P_t) - \min(P_t)} \quad (1)$$

3.2 Feature engineering

Feature engineering is applied to create temporal features using the input dataset. The timestamp is used to extract more detailed time features, which are required to enhance the prediction's explainability. The time-based features are extracted by converting the timestamp column into Date-Time format. The feature extraction is performed by following a step-by-step procedure to conduct an ablation study to add or remove relevant features from the dataset. Electricity consumption varies throughout the year, and the trend changes every season; therefore, a *month* feature is added. Different days of the week and days of the year consume different energy, so *day-of-week* and *day-of-year* features are created. Lastly *week-of-year* and *days-of-year* features are added. Energy consumption of dwellings is highly correlated with weather conditions and days. The correlation between energy consumption and weather conditions has been computed to determine relevant weather features.

3.2.1 Correlation analysis

Spearman's correlation coefficient is applied to determine the predictive relation between the energy consumption of home appliances and weather conditions. The correlation coefficient has been calculated to select the relevant features of the weather dataset that influence the amount of energy consumed by households. In the winter season, more energy is consumed by heating devices, but in the summer,

air conditioner usage is high. The correlation coefficient is calculated as follows in equation 2:

$$\text{corr}_\rho = \frac{1 - 6 \sum w_i^2}{n(n^2 - 1)} \quad (2)$$

where w_i is the difference between the ranks of corresponding weather variables, n is the total number of measurements. The sign of Spearman's correlation coefficient denotes the association between energy consumption (P) and weather features (Temp, humidity, wind_speed, pressure). Let's assume energy consumption as dependent variable (X) and weather features as independent variable (Y). If variable X tends to increase with the increase in variable Y then the correlation coefficient is positive, but if variable X tends to decrease with the increase in variable Y then the correlation coefficient is negative. When the dependent and independent variables are related monotonically, the correlation coefficient values are 1.

Using the real-time dataset, the correlation analysis in Fig. 2 depicts corr_ρ value 0.47 for P and Temp, -0.3 for P and humidity, 0.2 for P and wind_speed, -0.35 for P and pressure. The correlation coefficient for the IBLEND dataset obtained a value 0.02 for the power and Temp, -0.04 for power and humidity, -0.1 for power and wind_speed, and 0.07 for power and pressure. Hence, it can be validated that energy consumption has a positive correlation with temperature and wind_speed but a negative correlation with humidity

and pressure. While wind speed has a low correlation with energy consumption. Subsequently, the time-based features and climate features are further used to analyze their impact on electricity demand in residential buildings.

3.3 Electricity prediction and optimization approach

The main aim of this paper is electricity demand prediction in households, and the workflow of the proposed approach is illustrated in Fig. 3. This module is implemented using LSTM model and a genetic algorithm. LSTM is capable of learning from historical time steps, and it is widely used for solving time-series prediction problems. LSTM is a kind of recurrent neural network (RNN) that takes feature sequences as input, but RNN can not handle long-term dependencies in the sequential data. To overcome the limitations of RNN model, LSTM is equipped with storage cells and gates (Karijadi and Chou 2022). The LSTM model architecture comprises four primary components: input/output gate, cell, and forget gate (Yang and Wang 2022). Firstly, the forget gate f_t activation function is computed using input P_t by dropping the useless information in the previous cell state C_{t-1} . The forget gate equation 3 is given below (Yang and Wang 2022):

$$f_t = \sigma(W_f(h_{t-1} + P_t) + b_f) \quad (3)$$

It gives a value in the range of 0 and 1; a value close to 1 reveals the need to retain the information, and the value 0 means to remove the information. The cell state C_t is updated

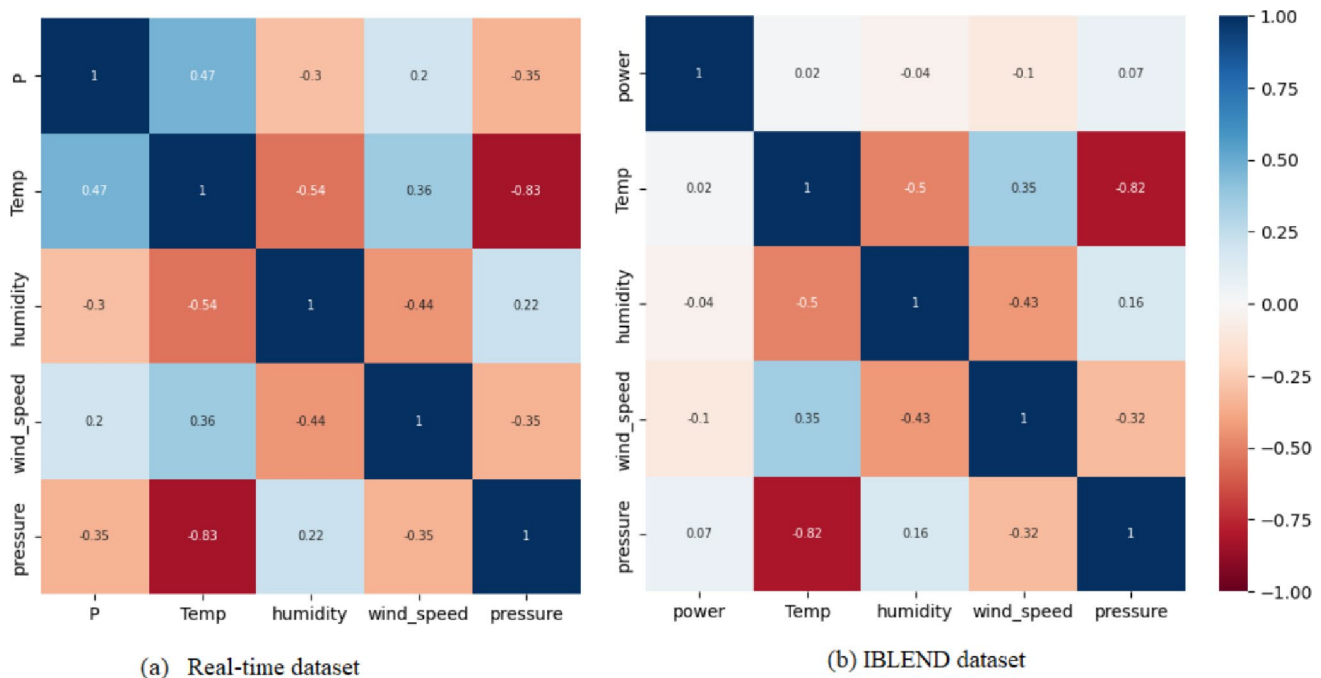


Fig. 2 Correlation analysis of energy consumption with the weather conditions (using real-time dataset and IBLEND dataset)

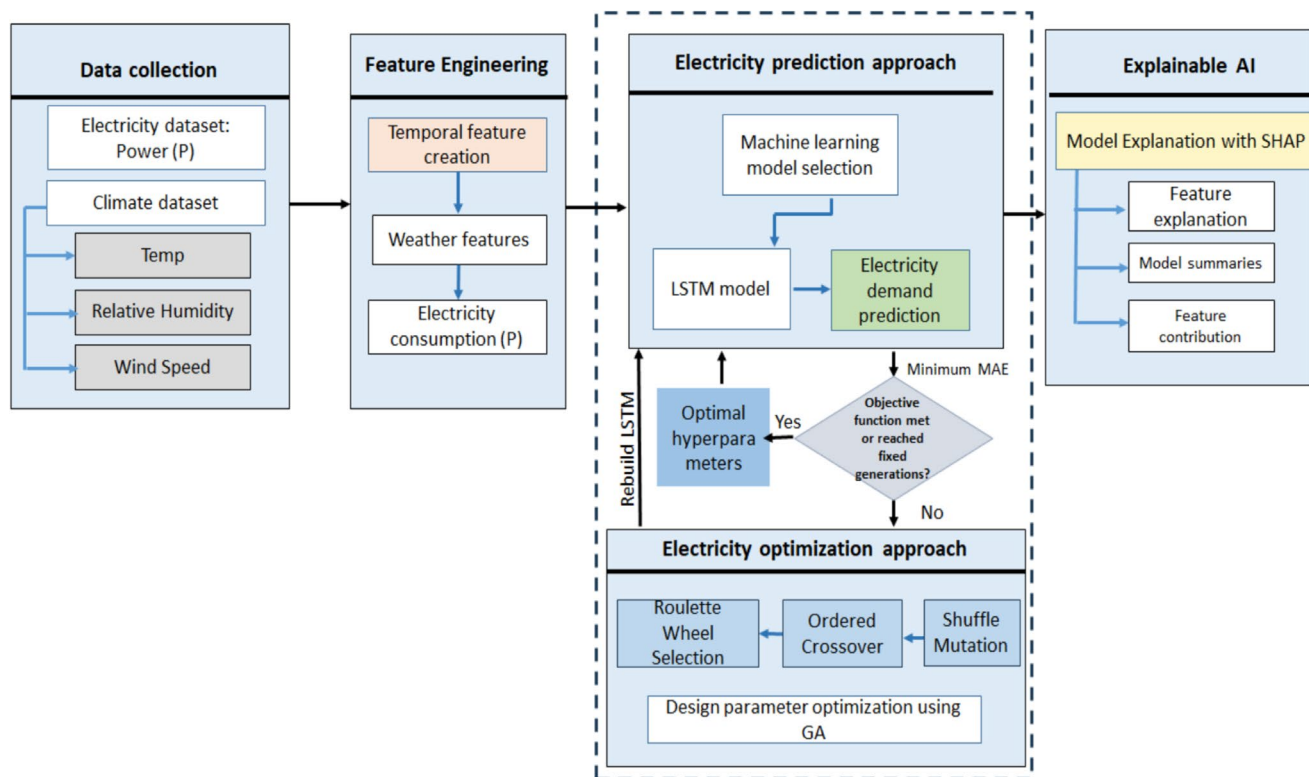


Fig. 3 Workflow of the proposed explainable AI-based electricity prediction and optimization approach for residential buildings

and produced using a vector of new candidate values $\tilde{C}_t$. The function of the cell is to choose the information for retaining for a long time, and the formula is given by Eqs. 4 and 5. The input gate i_t selects the information for memory cell at the timestamp i , given below in equation 6 (Karjadi and Chou 2022; Yang and Wang 2022):

$$\tilde{C}_t = \tanh(W_c(h_{t-1} + P_t) + b_c) \quad (4)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (5)$$

$$i_t = \sigma(W_i(h_{t-1} + P_t) + b_i) \quad (6)$$

The output o_t is calculated by multiplying the output value and updated cell state. To normalize the values between (-1) and (1) , the sigmoid layer filters the output, and then the modified cell state C_t is passed to $\tanh$. The output function is described in Eq. 7, and the revised hidden state is determined by Eq. 8 (Yang and Wang 2022).

$$o_t = \sigma(W_{oh}[\vec{h}_t, \vec{h}_t] + b_o) \quad (7)$$

$$h_t = o_t * \text{ReLU}(C_t) \quad (8)$$

Here P_t denotes the input while o_t is the output at time step t . The weights of vectors (W_i, W_f, W_c, W_o) show the input

gate, forget gate, cell state, and output gate and (b_i, b_f, b_c) represent the corresponding bias values. Before training the LSTM model, certain hyperparameters, such as input window size, hidden layers, neuron units, dropping rate, and learning rate, must be determined. Achieving optimal building hyperparameters using the trial and error method is an exhausting and time-consuming task. To address this concern, GA has been applied to obtain the optimal design parameters for the LSTM model to reduce the error and improve the prediction accuracy.

The Algorithm 1 provides the step-by-step procedure to select appropriate hyperparameters for LSTM model. GA is a heuristic search technique used to solve optimization problems and imitates the natural selection process. Taking different population sizes helps to fine-tune the genetic algorithm's parameters to achieve the best hyperparameters for LSTM. Initially, a relatively small population size (N) is taken and incremented gradually to larger population sizes like 20, 25, 30, 50, or 100. Finally, a population of chromosomes in the solution space is selected as 25, the number of generations as 10, and the gene length is assumed as 15. In this paper, along with LSTM layers, the transformer layer has been combined to handle the long-term time dependencies in the time-series data. The LSTM layer processes the daily and weekly consumption patterns, while the transformer layer deals with the seasonal variation in electricity

consumption data. The output of the combined GA-LSTM with the transformer model is estimated electricity consumption at the specific Timestamp. The training error (Mean Absolute Error (MAE)) of LSTM model is the objective function (acts as fitness criteria) that evaluates the quality of chromosomes to determine the probability of getting selected in the next generation. There are three operations, selection, crossover, and mutation, to generate the new chromosomes. Using the fitness score (MAE value), these three operations update the initial population with new offspring obtained by crossover and mutation. This procedure is repeated until the fittest offspring are obtained as the optimal solution.

Algorithm 1 GA-based hyperparameter optimization of LSTM model

Input: Electricity dataset (E)
Output: Optimal hyperparameters

- 1: Initialize GA population $N = 25$, $generations = 15$, $gene_length = 15$
- 2: Initialize LSTM hyperparameters randomly
- 3: *apply sliding_window approach*
- 4: *train and test split into 70 : 30 ratio*
- 5: *train LSTM – Transformer model*
- 6: *Evaluate fitness_score $\leftarrow$ MAE*
- 7: **for** *iteration* ≤ 70 **do**
- 8: *Selection: Train LSTM for fitness score*
- 9: *Crossover*
- 10: *Mutation*
- 11: *Test minimum fitness score*
- 12: **end for**
- 13: *model.predict(test)*
- 14: *Calculate loss = MAE*

The GA optimization approach finds the best hyperparameters, such as input window size, neuron units, hidden layers, and learning rate. For the LSTM model, input data needs to be transformed into a three-dimensional array in which the first dimension is input data size, the next is timestamps that can go back through time, and the last is the number of features. The sliding window approach transforms the time series data into supervised learning data by adding lag observations of input features. While training the model, the sliding window scans the entire dataset, and this process is iterated until the end of the dataset. The results of the electricity predictions are explained and interpreted using XAI tools in the subsequent section.

3.4 Explainable AI

This paper aims to apply the XAI model to households' electricity prediction. An explainable AI model is helpful

in interpreting the reason behind prediction results. It makes the decision-making process more user-centric and reasonable by providing trust, reliability, and understandability. In the field of electricity forecasting, XAI tools have been applied to investigate influential factors in electricity consumption. To interpret and explain the “black-box” ML model, the SHAP XAI tool has been integrated with the LSTM model. SHAP is based on the game theory approach, which fairly distributes the contribution of each player (Lundberg and Lee 2017). For the electricity prediction model, Shapley values are computed to quantify the contribution of each feature (Henriksen et al. 2022). The explainable SHAP model (Li 2022) can be defined by Eq. 9 as follows:

$$\phi(f, x) = \sum_{z_i \subseteq N \setminus \{i\}} \frac{k!(z_i - k - 1)!}{z_i!} (f_x(z_i) - f_x(z_i/i)) \quad (9)$$

The aim is to formulate SHAP as an ML problem to examine the impact of different features on the model output. According to Eq. 9, the Shapley value $\phi(f, x)$ of a feature is the overall contribution to the prediction model with various feature combinations, where f is a prediction model, z_i is the total number of features, k is a subset of input features, and x is a feature vector. The first half of the equation exhibits the weight that integrates the impact of adding or subtracting different features among all features z_i . The other half $(f_x(z_i) - f_x(z_i/i))$ represent the feature contribution. SHAP is implemented in Python, which provides visualization tools to depict the feature explanation. SHAP provides a separate explainable method for deep learning models, such as convolutional neural networks (CNN) and recurrent neural networks (RNN). In this paper, Deepexplainer is deployed to

Table 2 List of best hyperparameters chosen by GA optimization algorithm for electricity consumption datasets

Hyperparameters	Description	Value/range	
		Real-time dataset	I-BLEND dataset
Input window size	lag window calculated using genetic algorithm	14	21
Size of Output window	prediction horizon	1	1
Number of hidden layers	LSTM layers determined using genetic algorithm	2	3
Transformer layer	Transformer layer computes long-term seasonal dependencies	1	1
No. of neuron units	number of neuron units calculated using genetic algorithm	30	50
Learning rate	adjust weights of neural network	0.01	0.01
Epochs	number of epochs used during model training	50–70	70–100

explain GA-LSTM-based electricity prediction model (Manthana et al. 2023). After building and training the electricity prediction model, SHAP's Deepexplainer is deployed on the testing dataset despite training data due to its high computational requirement.

4 Experimental results and discussion

The proposed GA-LSTM approach has been implemented in Python Keras with TensorFlow as the backend, along with the XAI tool using the SHAP package. The performance evaluation used traditional performance metrics, namely, Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and MAE.

4.1 Case study 1: predictive analysis

The optimal set of hyperparameters computed using GA optimization are given by the Table 2, including the number of hidden layers, neuron units, learning rate, and input window size. Along with this, the number of epochs and output window size are decided during model training. For both the electricity consumption datasets, the learning rate

is the same but with different hidden layers, neuron units, and window sizes. It signifies that the LSTM model structure must be dynamic and adaptive for heterogeneous features of electricity consumption.

The proposed XAI-based hybrid GA-LSTM model has been evaluated using the household's real-time electricity usage dataset. To identify the influence of climatic conditions on electricity consumption, weather features such as temperature, humidity, wind speed, and pressure are merged into the electricity consumption dataset. In addition, electricity consumption is influenced by different days or months of the year; therefore, temporal features have been created by applying feature engineering. The electricity demand is predicted using the GA-LSTM model, and the model capability is assessed using 70% training and 30% testing data. K-fold cross-validation has been applied to the proposed prediction model to get a robust model. During training, the dataset is split into k-folds using KFold. To illustrate the effectiveness of the proposed approach, it is compared with other state-of-the-art machine learning models (SVR, RF, RNN, LSTM and LSTM with Transformer layer). Table 3 presents the prediction results using MSE, RMSE, and MAE for the training and testing phase. The electricity predictions in the testing phase of real-time dataset and I-BLEND dataset are shown in Fig. 4, respectively. The x-axis depicts

Table 3 Performance results of various prediction models using real-time electricity dataset and I-BLEND dataset

Electricity dataset	Prediction model	Training results			Testing results		
		MAE	RMSE	MSE	MAE	RMSE	MSE
Real-time dataset	SVR	0.215	0.265	0.178	0.221	0.274	0.183
	RF	0.202	0.247	0.161	0.209	0.253	0.167
	RNN	0.189	0.231	0.149	0.191	0.239	0.152
	LSTM	0.142	0.167	0.139	0.154	0.178	0.144
	Hybrid GA-LSTM	0.051	0.026	0.047	0.0801	0.051	0.073
I-BLEND	SVR	0.244	0.276	0.178	0.259	0.282	0.186
	RF	0.193	0.242	0.158	0.203	0.253	0.165
	RNN	0.175	0.213	0.124	0.188	0.221	0.135
	LSTM	0.167	0.179	0.118	0.173	0.184	0.109
	Hybrid GA-LSTM	0.044	0.028	0.012	0.062	0.032	0.081

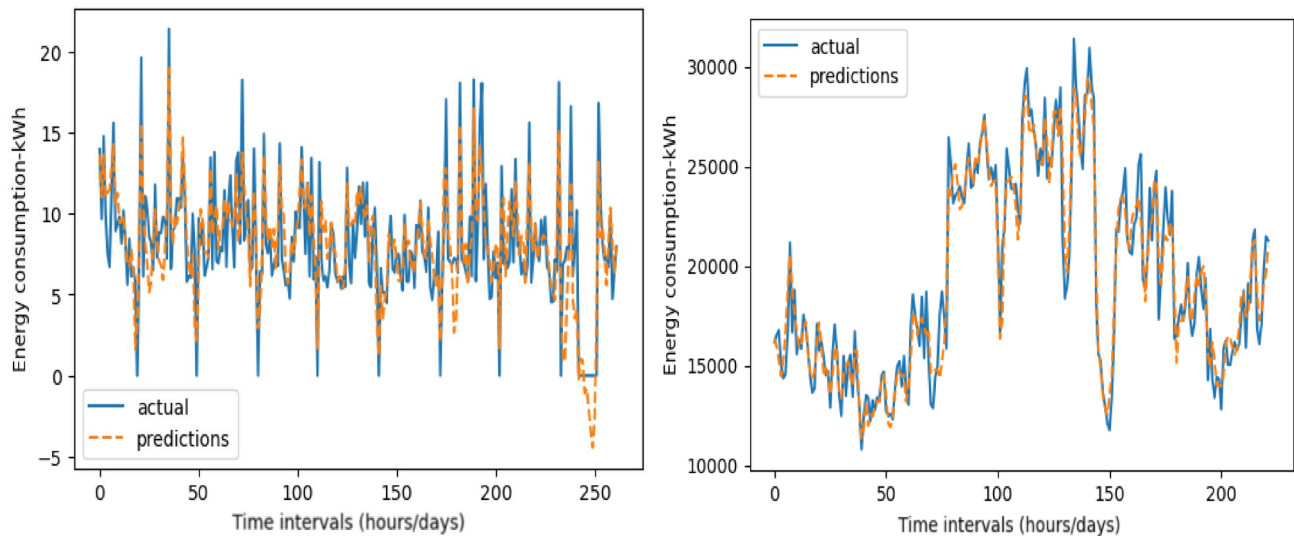


Fig. 4 Predictive performance of proposed explainable GA-LSTM model using real-time electricity consumption dataset and I-BLEND dataset

the number of data points and y-axis is daily energy consumption in building. The actual energy used is shown by blue lines, and dotted orange lines depict predicted values. As seen in Fig. 4, the predicted electricity consumption is close to the actual electricity consumption for both datasets, which exhibit dissimilar electricity consumption patterns. It is analyzed that harnessing the capability of GA for hyperparameter optimization refines the LSTM model architecture. Certain insights can be drawn from the prediction results given in Table 3, which are discussed below.

- The inclusion of climatic conditions and temporal parameters enhanced the accuracy of the LSTM model, attaining MAE (0.154 kWh and 0.173 kWh) as compared to the RNN model with MAE (0.191 kWh and 0.188 kWh).
- By incorporating GA optimization technique with LSTM, the proposed GA-LSTM model outperforms the LSTM model for both real-time electricity dataset and I-BLEND dataset.
- During the testing phase, the proposed explainable AI-based GA-LSTM model attained MAE (0.0801 kWh), RMSE (0.051 kWh), and MSE (0.073 kWh) using a real-time electricity dataset. Using I-BLEND dataset, the proposed work obtained MAE (0.062 kWh), RMSE (0.032 kWh), and MSE (0.081 kWh).
- It is found that the suggested GA-LSTM approach has attained a minimum error rate as compared to the work of Gaur and Majumdar (2019), they had reported the prediction error (MAE 0.089 kWh and 0.067 kWh) using LSTM model on I-BLEND dataset.

4.2 Case study 2: explainability analysis

The aim of XAI tool is to improve the predictive performance using the information about the internal workings of GA-LSTM model. In this paper, the contribution of climatic conditions and temporal features to prediction results has been investigated using the SHAP method. It computes the Shapely values to determine the relationship between independent and dependent features such as electricity consumption, temperature, humidity, wind speed, pressure, and days of the month. The SHAP values can be used to interpret model behavior both at a local and global level.

4.2.1 Global explainability

The global explainability shows the overall feature contribution of the entire dataset. The SHAP summary plot offers a comprehensive visual representation of features influencing the prediction of energy consumption and the global interpretation of the whole dataset. This graph shows the aggregated Shapely values for all feature instances in the dataset, and these Shapely values provide the contribution of the feature, either positive or negative, in electricity prediction outcomes. The summary plot in Fig. 5 shows the SHAP values for the GA-LSTM model-based prediction model. The features with higher SHAP values have a higher influence on prediction results. The x-axis represents the SHAP values, and y-axis depicts the feature value.

The SHAP values of each feature are ranked from top to bottom, where the most important features are shown at the top of the summary plot, and the least important features are depicted at the bottom.

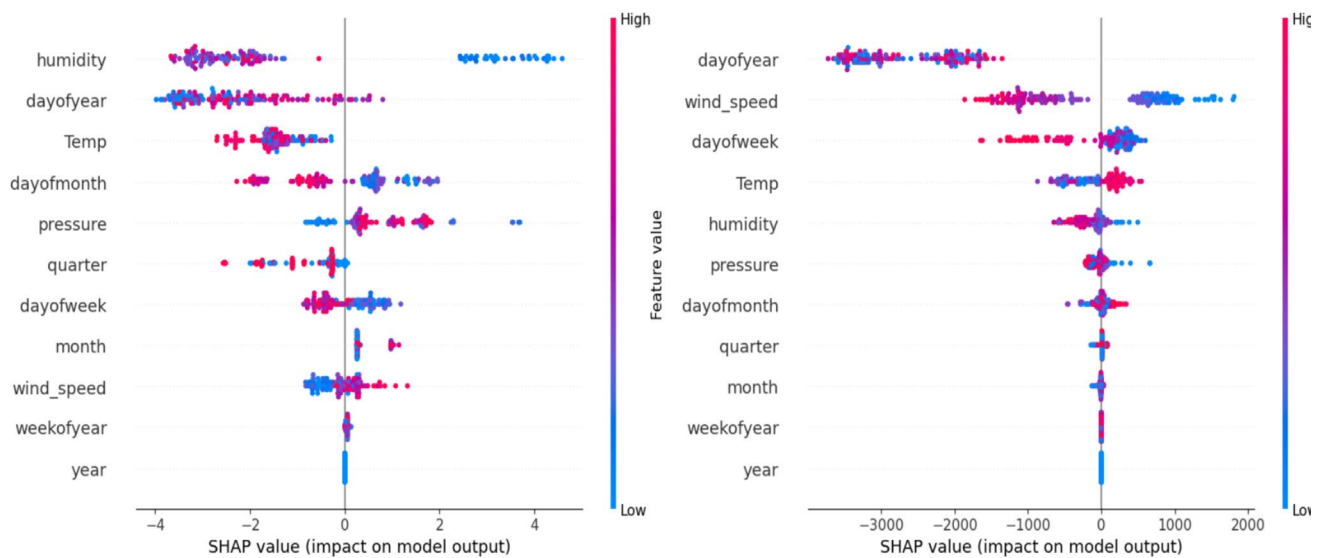


Fig. 5 SHAP summary plot for LSTM model on **a** real-time electricity consumption dataset and **b** I-BLEND dataset (where x-axis shows Shapley values and y-axis depicts the features)

The summary plot in Fig. 5a represents the prediction explanations using real-time electricity dataset. The weather features *humidity*, *day-of-year*, *Temp* and *day-of-month* are most important features while *pressure*, *quarter*, *day-of-week*, and *wind_speed* also contributes positively in prediction. The feature *month*, *week-of-year* and *year* has least contribution in electricity prediction. Weather features such *humidity*, *Temp* highly impact the heating and cooling requirement in buildings while *day-of-year* impacting the energy consumption patterns due to seasonal variation in energy demand. The second summary plot in Fig. 5b is

created using benchmark dataset, I-BLEND, in which *day-of-year*, *wind_speed*, and *day-of-week* are most important features. The other features with a positive influence on electricity predictions are *Temp*, *humidity*, *pressure*, and *day-of-month*. The least important features are *quarter*, *month*, *week-of-year* and *year*, shown at the bottom of summary plot. The feature importance values are helpful in estimating energy consumption during peak hours/days or during critical weather conditions and offer insights to consumers to optimize electricity usage at extreme times.

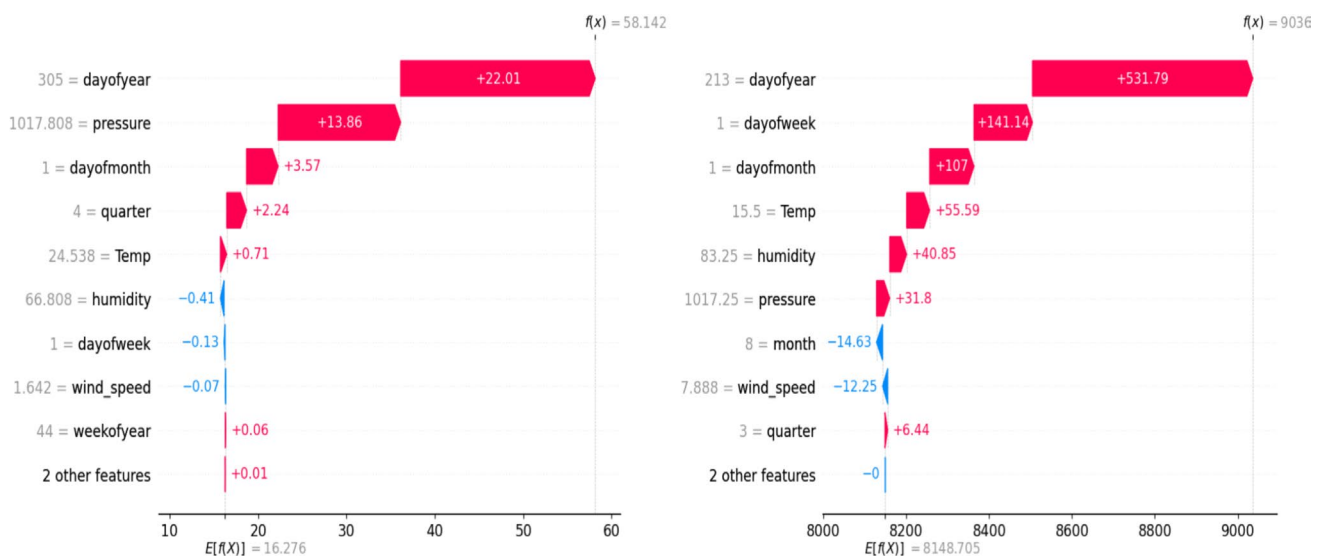


Fig. 6 SHAP waterfall plot for LSTM model on **a** real-time electricity consumption dataset and **b** I-BLEND dataset (where the x-axis depicts the predicted values and the y-axis represents the input features)

4.2.2 Local explainability

This section focuses on the local explanation for individual predictions rather than providing the overall summary of whole dataset predictions. These explanations offer local interpretation and clarify the reasoning behind every single prediction.

A waterfall plot is a local explainability method that breaks down the model's prediction into base value and individual feature contributions. The waterfall plot provides visual feature contributions to individual feature stacks employing the SHAP values as horizontal bars. The length of horizontal bars varies according to the SHAP value of each feature, and the SHAP values sort the features in ascending order. The plot is read from bottom to top, where the x-axis depicts the predicted values and the y-axis represents the input features. Figure 6a shows the waterfall plot of real-time electricity consumption dataset, in which *day-of-year* and *pressure* are most important features while *humidity*, *day-of-week*, *wind_speed*, *month*, and *year* are least contributing features. The predicted energy consumption for the real-time electricity dataset is 58.142 kWh relative to the baseline value of 16.276 kWh, while the final estimated energy demand for the I-BLEND dataset on a particular day is 9036.424 kWh, corresponding to the baseline value of 8148.705 kWh, which shows the average predicted energy consumption for the whole dataset. The second waterfall plot is made to explain the prediction of I-BLEND dataset as shown in Fig. 6b; in this case, the highest contributions are given by features *day-of-year*, and *day-of-week*. On the other hand, least important features are *month*, *wind_speed*, *week-of-year*, and *year*. For both datasets, *day-of-year* shows

a positive contribution in predictions due to repeating consumption patterns. Waterfall plots provide meaningful insights into the impact of climatic conditions and temporal features on individual energy predictions.

Force plots are used to intuitively observe the local interpretations within a short data frame. The force plot, as shown in Fig. 7, depicts how individual features influence the prediction performance. The numbers on the arrows are the feature values. The positive feature contributions are expressed as red arrows, and the negative feature contributions are depicted as blue arrows. The features shown by red arrows allow to increase the predicted values more than the base value. Subsequently, the blue arrows decrease the predicted values below the base values. The size of the arrow represents the importance of specific features in the prediction model; longer arrows have a greater influence on the prediction results. The force plot in Fig. 7a depicts a base value of 12.36 kWh which is the average predicted energy for the whole dataset, and Fig. 7b plot presents the base value of 19526.78 kWh mean of the prediction of the entire dataset. Figure 7a shows the force plot for a real-time dataset using a short data frame of 10 samples. The predicted values, along with feature contribution, are determined using the SHAP values. The features shown on red arrows *wind_speed*, *day-of-month* and *pressure*, push the predictive performance higher. Further, Fig. 7b illustrates the force plot for IBLEND dataset. The features *pressure*, *day-of-week*, *wind_speed* and *day-of-year* are contributing positively to enhance the prediction performance. Overall, the results reveal that *day-of-year* has highest contribution to final electricity prediction. *pressure* and *day-of-week* has significant contribution in the prediction results.

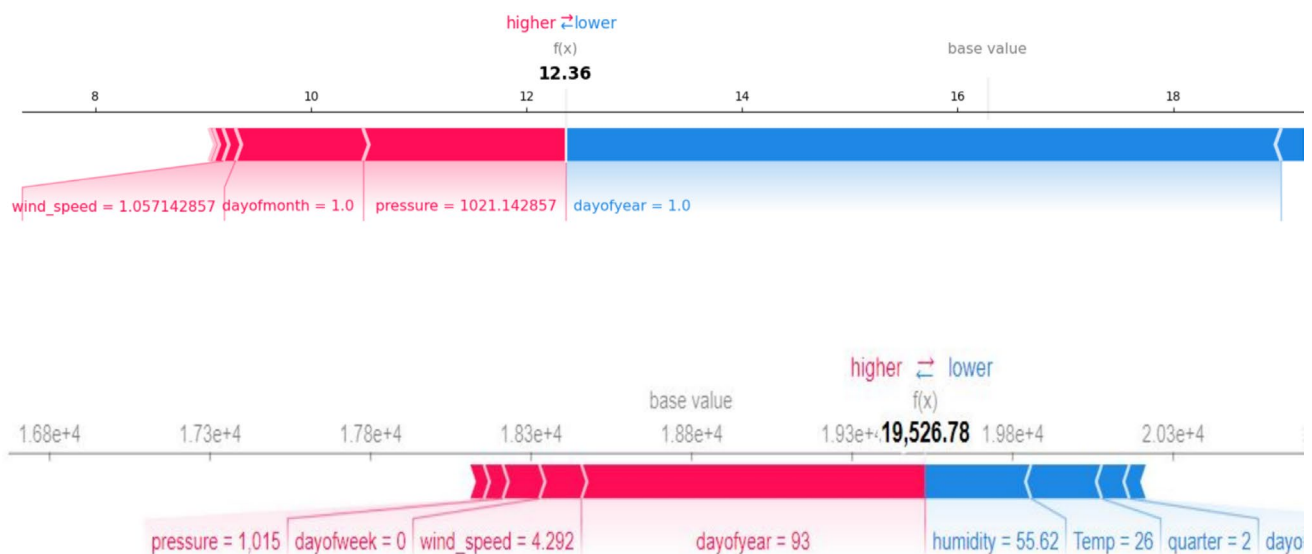


Fig. 7 SHAP force plot for LSTM model on **a** real-time electricity consumption dataset and **b** I-BLEND dataset

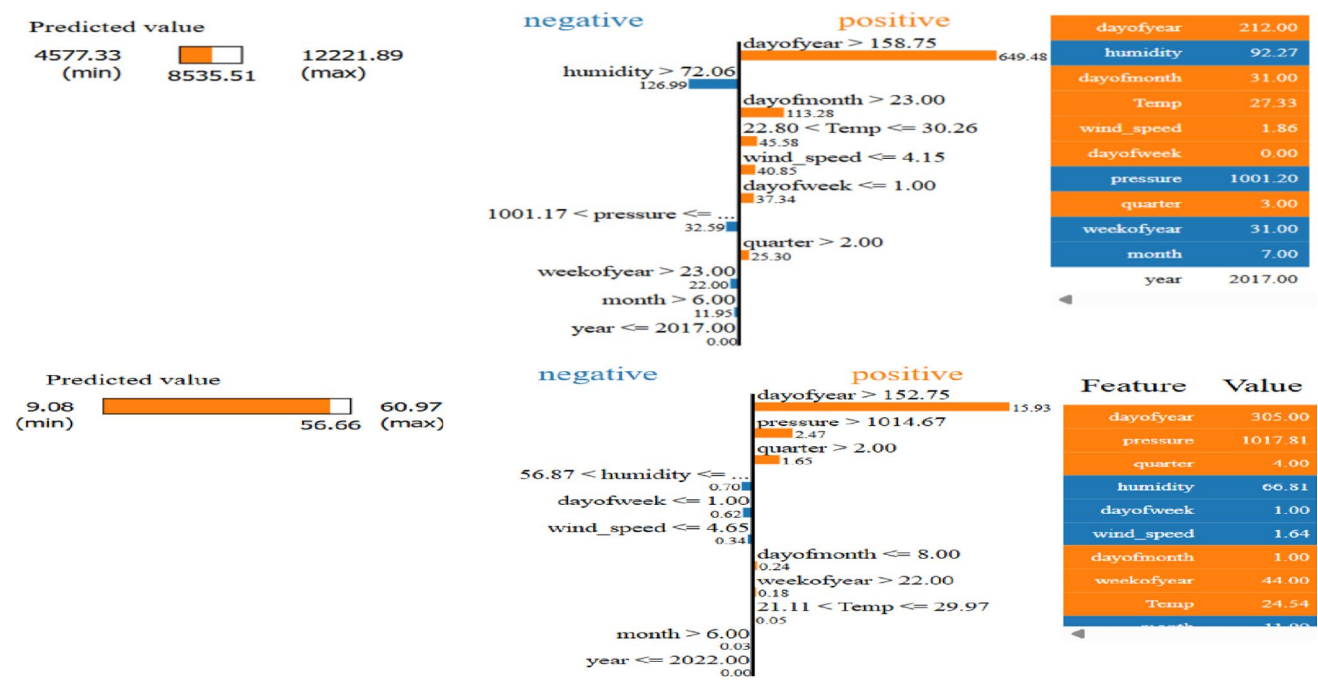


Fig. 8 LIME explanations for **a** real-time electricity consumption dataset and **b** I-BLEND dataset

Further, the local interpretability of individual electricity predictions has been assessed using LIME (Local Interpretable Model-agnostic Explanations). LIME provides an estimate of the behavior of complex models locally to explain individual predictions of deep learning models. The contribution of individual features, both positive and negative, is shown in Fig. 8. It depicts the impact of each feature on electricity prediction; the positive feature increases the prediction performance, while negative features decrease the prediction accuracy. The feature *day-of-year* contributed positive value to prediction and *humidity* have negative contribution in electricity prediction. The explanation graph indicates that the weather conditions and specific days of the year contribute to electricity consumption prediction.

4.3 Implications and limitations

The proposed prediction approach can benefit the electricity distribution sector and consumers by raising awareness through daily energy consumption patterns. It is expected that the analysis and results obtained through this work would help to save electricity for buildings with a realistic future forecast. The explainability analysis demonstrated the most important features that can be considered while forecasting the electricity demand in buildings. It also highlighted the least contributing features that can be discarded to get precise predictions. Incorporating real-time electricity data and weather conditions provides meaningful insights into real-world circumstances. Different types of residential

buildings and diverse geographical locations can be considered to verify the generalizability of the proposed approach. In some cases, the XAI tools have the limitation of capturing explanations for each prediction, which could result in imprecise understanding. XAI techniques have many applications in real-world scenarios, such as enhancing decision-making in automated systems, identifying influential parameters in renewable energy predictions, failure detection, and optimizing energy consumption in IoT-enabled intelligent buildings.

5 Conclusion

This research paper presents an explainable hybrid GA-LSTM model for electricity prediction in residential buildings. Initially, the optimization algorithm GA was integrated with the LSTM-transformer model to optimize its hyperparameters and get accurate predictions. Further, SHAP XAI tool has been adopted to interpret the predictions of LSTM model, as well as to analyze the impact of temporal and climatic features on electricity prediction. The performance of the proposed XAI-based GA-LSTM model has been evaluated using a real-time electricity consumption dataset and I-BLEND dataset. The results demonstrate that the proposed smart electricity prediction approach has attained the lowest prediction error and optimized the MAE by 14% (approx.) compared to the existing LSTM model for the I-BLEND dataset. This research work concluded that the GA has the

potential to optimize the prediction accuracy of LSTM model, and XAI tools are helpful in interpreting the reasoning behind individual electricity predictions.

6 Future scope

In the future, the proposed approach can be extended to analyze the application of the proposed XAI-based prediction model, considering different types of buildings to enhance their robustness. The explainable AI-based prediction model can be applied to datasets of various regions, like urban, rural, and different climate zones, to evaluate its adaptability. The insights provided by XAI approach can be used to design energy efficiency policies and sustainable energy systems for buildings. Hybrid models could be developed by combining the strengths of LSTM, CNN, GRU, XGBoost, and RF to improve the predictive performance. Advanced transfer learning architectures, such as temporal fusion transformer and informer, can be integrated to enhance prediction frameworks.

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Data availability The I-BLEND dataset that supports the findings of this research is available openly in the figshare Repository at <https://doi.org/10.6084/m9.figshare.c.3893581>, and the second dataset is available on reasonable request.

Declarations

Conflict of interest The authors declare no Conflict of interest in preparing this article.

Ethical approval This article does not contain any data on human participants or animals performed by any authors.

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